Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/06/2014
PI Name:	Ebert, Douglas Ph.D.		
Project Title:	Analysis of a Clinical Database as a Means to Validate Non-invasive Assessment of Intracranial Pressure Using the Cerebral and Cochlear Fluid Pressure (CCFP) Analyzer		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	06/13/2012	**End Date:**	11/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date extended to 11/30/2013 (original end date was 6/30/2013 and was later extended to 9/30/13) per PI (Ed., 10/25/13) NOTE: End date is now 9/30/2013 per PI (Ed., 6/19/13)		
Key Personnel Changes/Previous PI:	James Fiedler was added as Co-I for his contribution with analysis software.		
COI Name (Institution):	Danielson, Dick Ph.D. (Baylor College of Medicine) Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association) Gianoli, Gerard M.D. (Ear and Balance Institute) Soileau, James M.D. (Ear and Balance Institute) Fiedler, James Ph.D. (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
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Task Description:	Introduction: Visual impairment / intracranial pressure (VIIP) is a top risk for human spaceflight, yet there are no reliable means to monitor intracranial pressure (ICP) noninvasively, particularly inflight. A lumbar puncture (LP) procedure is invasive and thus not a viable option for inflight monitoring. Development and validation of a noninvasive means of monitoring ICP would result in collection of data that could define the involvement of elevated ICP in the VIIP syndrome. The Cochlear and Cerebral Fluid Pressure (CCFP) analyzer (Marchbanks Measurement Systems, UK) measures tympanic membrane displacement (TMD) in both evoked and passive modes. These TMD waveforms change in response to ICP alterations, allowing a noninvasive means to monitor ICP changes. Additional clinical validation of the CCFP analyzer is necessary to better understand the device's utility. The Ear and Balance Institute (Covington, LA) has over 1,000 patient records of CCFP tests, many with accompanying LP measures. Analysis of the results of these tests has helped validate the use of the CCFP technology as a surrogate measure of ICP. The purpose of this retrospective clinical study was to examine this unique CCFP and LP data set, in order to identify characteristics that could contribute to follow-on studies employing the CCFP analyzer. Methods: Charts selected for inclusion in the data set had CCFP and LP measures obtained within several days of each other, with no medical interventions between measures. The analysis had several components that addressed the specific aims of this study: 1) evaluate both evoked and passive TMD waveforms as measured by CCFP and compare with LP data; and 2) determine the age-dependent distribution of cochlear aqueduct patency. Discussion: Because the CCFP had been used as a clinical screening tool, many patients had begun medical intervention between administration of the CCFP test and the LP, or had been undergoing treatment prior to any test being administered. Another caveat associated with this analysis is that CCFP measures had not been collected at the same time as LP. To address these issues, our analysis focused on patients whose LP was performed in close temporal proximity to CCFP (within an average of ~5 days), and those who had no interventions between measures. These factors have reduced the number of charts available for analysis (~70 out of ~1000 available with CCFP and LP measures). Nearly all patients in this database had ear-related pathology, with the largest contributors being superior semicircular canal dehiscence (SSCD) (~50%), and perilymph fistula (~65%) that likely affect CCFP response. Despite these factors, a negative correlation between tympanic membrane displacement in an evoked CCFP test (Vm) and LP opening pressure was found in the non-SSCD patient population in the seated position ($p<0.002$), and expectation-maximization estimates suggested that Vm in the supine position trends negatively with LP ($p=0.068$). These results are encouraging as to the potential utility of the CCFP as an ICP screening tool in a non-clinical population, and deserve further investigation in an experimental setting.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research and there is insufficient time for solicitation. There is an immediate need for the non-invasive assessment of intracranial pressure (ICP). The CCFP provides a potentially viable option for monitoring ICP terrestrially and in-flight; flight units have been designed and assembled for a prior Shuttle DTO, greatly reducing the time required to develop the instrument for flight. However, additional clinical validation of this device is necessary in order to better understand the device's utility. The Ear and Balance Institute has a large clinical CCFP database that may provide data to help with validation and confidence in the potential that this technology offers as a surrogate measure of ICP. Results of this study will inform follow-on studies employing CCFP.
Research Impact/Earth Benefits:	This research has contributed to the knowledge and understanding of a non-invasive means of monitoring intracranial pressure. If fully validated, this methodology could provide terrestrial patients with a rapid means of monitoring intracranial pressure that would not require extensive training or invasive procedures.
Task Progress:	This task is complete--all data has been collected and analyzed, and the following abstract summarizes the work: Introduction: Visual impairment / intracranial pressure (VIIP) is a top risk for human spaceflight, yet there are no reliable means to monitor intracranial pressure (ICP) noninvasively, particularly inflight. A lumbar puncture (LP) procedure is invasive, thus not a viable option for inflight monitoring. Development and validation of a noninvasive means of monitoring ICP would result in collection of data that could define the involvement of elevated ICP in the VIIP syndrome. The Cochlear and Cerebral Fluid Pressure (CCFP) analyzer (Marchbanks Measurement Systems, UK) measures tympanic membrane displacement (TMD) in both evoked and passive modes. These TMD waveforms change in response to ICP alterations, allowing a noninvasive means to monitor ICP changes. Additional clinical validation of the CCFP analyzer is necessary to better understand the device's utility. The Ear and Balance Institute (Covington, LA) has over 1,000 patient records of CCFP tests, many with accompanying LP measures. Analysis of the results of these tests has helped validate the use of the CCFP technology as a surrogate measure of ICP. The purpose of this retrospective clinical study was to examine this unique CCFP and LP data set, in order to identify characteristics that could contribute to follow-on studies employing the CCFP analyzer. Methods: Charts selected for inclusion in the data set had CCFP and LP measures obtained within several days of each other, with no medical interventions between measures. The analysis had several components that addressed the specific aims of this study: 1) evaluate both evoked and passive TMD waveforms as measured by CCFP and compare with LP data; and 2) determine the age-dependent distribution of cochlear aqueduct patency. Discussion: Because the CCFP had been used as a clinical screening tool, many patients had begun medical intervention between administration of the CCFP test and the LP, or had been undergoing treatment prior to any test being administered. Another caveat associated with this analysis is that CCFP measures had not been collected at the same time as LP. To address these issues, our analysis focused on patients whose LP was performed in close temporal proximity to CCFP (within an average of ~5 days), and those who had no interventions between measures. These factors have reduced the number of charts available for analysis (~70 out of ~1000 available with CCFP and LP measures). Nearly all patients in this database had ear-related pathology, with the largest contributors being superior semicircular canal dehiscence (SSCD) (~50%), and perilymph fistula (~65%) that likely affect CCFP response. Despite these factors, a negative correlation between tympanic membrane displacement in an evoked CCFP test (Vm) and LP opening pressure was found in the non-SSCD patient population in the seated position ($p<0.002$), and expectation-maximization estimates suggested that Vm in the supine position trends negatively with LP ($p=0.068$). These results are encouraging as to the potential utility of the CCFP as an ICP screening tool in a non-clinical population, and deserve further investigation in an experimental setting.

Bibliography Type:	Description: (Last Updated: 03/09/2026)
Abstracts for Journals and Proceedings	Ebert D, Gianoli G, Soileau J, Ploutz-Snyder R, Danielson R. "Analysis of clinical records as a means to validate non-invasive assessment of intracranial pressure using the Cerebral and Cochlear Fluid pressure (CCFP) analyzer." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3287.pdf , Feb-2014
Abstracts for Journals and Proceedings	Ebert D, Gianoli G, Soileau J, Ploutz-Snyder R, Danielson R, Hart S. "Analysis of clinical records as a means to validate non-invasive assessment of intracranial pressure using the Cerebral and Cochlear Fluid pressure (CCFP) analyzer." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013